YOLOv8-AFPN: Asymptotic Feature Pyramid Network for Real-Time Object Detection

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Abstract- *Object detection in complex urban scenes requires high representational fidelity and minimal latency. We present an improved YOLOv8 model.*

I. INTRODUCTION

Modern computer vision models have experienced dramatic evolution. Developing architectures that balance latency and feature selectivity remains a central topic of inquiry in machine intelligence.

II. RELATED WORK

Prior approaches often struggled with dense multi-scale object distributions.

III. PROPOSED METHODOLOGY

In this section, we delineate our proposed framework. The core system architecture employs YOLOv8 with Asymptotic Feature Pyramids. We integrate asymptotic multi-level feature pyramid routing, connecting cross-layer feature maps with learned convolutional kernels. Training is executed over the MS COCO benchmark. The loss formulation balances bounding-box focal regression with cross-entropy classification.

IV. EXPERIMENTAL RESULTS

To evaluate the proposed model, extensive experiments were conducted on MS COCO. The proposed architecture achieved 94.8% mAP accuracy. Comparative evaluations demonstrate substantial improvements over previous YOLO iterations on small objects.

V. CONCLUSION

In this work, we introduced a novel framework that improves upon established baselines.